

Tree of Thought: A Design Decision

Table for Four. CMU AI Agent Certification, Capstone, Week 4.
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Week 4 asks whether Tree of Thought reasoning belongs in the capstone. It does not, and what follows is the argument for that conclusion rather than a shrug at the technique. Section 2 works out what a Tree of Thought implementation would concretely have looked like in this architecture. Section 3 shows why each part of it loses to what is already built. Section 4 takes the single strongest candidate site and shows the decision even there. Sections 5 to 7 give the reasoning approach that does fit, the risk that choice accepts, and the conditions under which the decision should be reopened.

1. Verdict

Tree of Thought reasoning is not appropriate for Table for Four, and adopting it would hinder the design rather than improve it.

The short form of the argument runs in three parts. The alternatives this agent must weigh are retrieved from the world rather than generated by the model, so the breadth of any tree would be manufactured rather than real. The selection among those alternatives is deliberately the guest's, so an internal search that returns a single answer would delete the central promise of the product. And the constraints that actually matter are settled by tool calls with exact answers, so a scored critic would be a weaker instrument than the function it replaced.

2. What Tree of Thought would have looked like here

Rejecting a technique in the abstract is not a design decision, so the mechanics are worth stating concretely. Mapping the vocabulary of Tree of Thought onto this system gives a coherent reading.

Concept	Nearest thing in Table for Four
Thought	One candidate outing: a restaurant, a date, a time, a party size and any perk, partially filled in.
Node	The state of that partial outing once one further constraint has been resolved.
Branch	An alternative resolution: a different restaurant, a different time slot, or a different constraint relaxed.
Depth	The stages of constraint resolution, from cuisine and area, to restaurant, to time, to confirmation.
Branching factor	About four restaurants, since the shortlist is capped at four, and then three to six open times for whichever one is chosen.

Depth limit	Three or four, because the outing is fully specified by that point.
Termination	A complete outing that passes an availability check and receives the guest's approval.

The tree is therefore easy to imagine, and that is exactly what makes the rejection worth arguing rather than asserting. The tree is already there. The real question is not whether this problem has a search structure. It is who runs the search, and what the evaluator is made of.

3. Why Tree of Thought loses on this problem

3.1 The branches are retrieved, not reasoned

Tree of Thought earns its cost when the solution space has to be invented by the model: proofs, puzzles, program synthesis, open ended planning, cases where the model can produce a genuinely novel next step that no external system could have handed it.

In this system the candidate set arrives from the Google Places API through the search server, already filtered by cuisine, area, price and rating. The model does not think its way to a fifth restaurant. It fetches one. Asking a language model to expand branches over a retrieved list adds a generation step whose output is bounded by data the model was already given. That is not reasoning. It is restating the list.

3.2 The human is the search algorithm, by design

This is the decisive point.

The premise of the product, stated in the scoping document and enforced in code, is that the guest chooses. The concierge offers a shortlist and the guest picks the restaurant. The concierge offers the open times and the guest picks the time. The concierge reads the booking back and waits for a yes. Standing preferences, meaning the guest's home area, their usual party size and their favourite cuisines, cannot be changed at all without the guest's explicit agreement.

So the tree already runs. Breadth four, depth three, and the guest is the critic.

Tree of Thought is a technique for letting the model explore alternatives instead of asking. Internalising this search would mean scoring four restaurants against an invented rubric, pruning three of them, and presenting the survivor as though it were the answer. For a hospitality product whose entire differentiator is that it asks rather than assumes, that is not a neutral architectural choice. It is the precise failure mode the system was built to prevent. The consent gate that protects standing preferences would then sit next to a search procedure designed to make the same class of unilateral judgement the gate exists to forbid.

3.3 There is no trustworthy evaluation signal at conversation time

Tree of Thought needs a state evaluator good enough that pruning on its score beats not pruning. It is worth asking what the critic would actually be scoring here. Is one Italian restaurant a better dinner than another? There is no ground truth. Rating and price level are already retrieved and already shown to the guest. Everything beyond them, meaning how well the room suits the occasion, whether this particular birthday wants noise or quiet, how a partner feels about the neighbourhood, is precisely the knowledge the guest holds and the model does not.

Where this project can measure retrieval quality, it does so offline and properly. The perks retriever is scored against a labelled benchmark using hit at k, precision at k, and mean reciprocal rank, deterministically and repeatably. That is the right shape for an evaluation signal: fixed labels, standard information retrieval metrics, and a result that does not drift between runs.

A language model critic scoring restaurant desirability during a live conversation would have none of those properties. Weak evaluation signals leading to the pruning of the best branch is usually listed as a risk to be mitigated. Here it is not a risk to mitigate. It is disqualifying, because the branches being pruned are real restaurants that the guest then never sees, discarded on a score with nothing behind it.

3.4 The constraints are checked, not judged

Every hard constraint in this domain has an exact answer available from a tool call. Is the table free at seven in the evening. Does the party size fit. Is the booking more than 24 hours away, which is the cancellation policy. Is an email on file before booking. Does this perk apply on a Friday for four people.

None of these is improved by a scored branch. A constraint with a deterministic checker should be checked rather than estimated, because the check is cheaper, exact, auditable, and cannot hallucinate. The constraint complexity argument for Tree of Thought applies where satisfaction is expensive to test or must be judged. Here it is a function call.

3.5 Cost and latency, measured against a live demonstration

A single guest turn currently costs between one and seven model calls. A modest tree at breadth three and depth three adds roughly a dozen generations plus a dozen evaluations at every decision point. In a chat interface being screen recorded for the capstone demonstration, that is a visible stall in the middle of a conversation, paid on every outing, in order to arrive at a restaurant the guest was going to be shown in any case.

4. The strongest candidate site, and why it still loses

Steelmanned honestly, one place in this architecture does have a genuine search structure: constraint relaxation after a search returns nothing. A request such as vegan friendly with a gluten free option, a high chair, Friday at seven, in Midtown, moderately priced, and carrying a

perk, can easily come back empty, and something then has to give. Which constraint to relax is a real branching decision with real consequences.

That decision is already solved in the orchestration graph by a fixed relaxation ladder, ordered by hospitality knowledge: first drop the minimum rating, then raise the price ceiling by one band, then drop the filter for places open now, then drop the cuisine filter. The loop is bounded at two iterations and routed by a conditional edge.

Three reasons the ladder beats a tree even here.

- **The evaluator is a tool call rather than a critic.** Whether the search returned candidates is answered exactly, by the search itself. Tree search buys the ability to compare branches under uncertainty, and there is no uncertainty here to price.
- **The branching factor is about four, with an obvious ordering.** A guest would rather eat at a restaurant rated 4.1 than not eat. They would rather pay one price band more than abandon dinner. They would rather change cuisine than change nothing. That ordering is stable hospitality knowledge, not something worth rediscovering per query at inference time.
- **It costs no model calls and stays auditable.** Each relaxation writes its reason into the state log, which feeds the governance trail planned for the fourth milestone. A tree expansion would produce a decision that no log could faithfully reconstruct.

When a fixed policy reaches the same answer as a search, at no model cost, with a readable audit line, the search is not the better engineering.

5. The reasoning approach that fits

Grounded ReAct, with a bounded refinement loop and human selection at the branch points.

The framing deserves one correction. The usual alternative offered to Tree of Thought is a linear chain of thought, and this project uses neither. It is not a single linear chain, because breadth is genuinely present. That breadth is simply externalised to the guest rather than collapsed inside the model.

Four components, all built:

1. **A ReAct style tool loop.** The model observes, calls a tool, reads the result, and decides the next action, bounded at six tool hops per turn. Reasoning is interleaved with real observations rather than running ahead of them.
2. **Deterministic orchestration as a state machine.** The LangGraph pipeline runs parse, search, refine, match perks, rank, propose, gate, book, audit. Control flow is inspectable in LangGraph Studio and reproducible under test.

3. **Bounded backtracking.** The relaxation ladder of section 4: one constraint at a time, capped, and logged.
4. **Human selection at every branch point.** The shortlist, the time, the booking confirmation, and any change to a standing preference. The approval gate makes that approval explicit and recorded.

All of it rests on retrieval. Google Places supplies candidates, a hybrid semantic and metadata search supplies perks, and Tavily supplies menu highlights with citations. The job of the model is to converse over grounded facts, never to invent a restaurant, a time, or a dish.

Why this serves the goal better than a tree

	Tree of Thought	This architecture
Where branches originate	Model generation	Retrieval from real inventory
Who evaluates	A model critic against an invented rubric	The guest, against their own occasion
How pruning happens	A score threshold, silently	The guest declines, visibly
Termination	Depth limit or critic satisfaction	The guest says yes and a confirmation id is returned
Auditability	Discarded branches are hard to reconstruct	Every step is logged for the governance trail
Cost per decision	Roughly ten to twenty five extra model calls	No extra model calls
Worst failure	Prunes the restaurant the guest actually wanted	The guest sees the option and says no

The goal of this agent is not to compute the optimal restaurant. It is to help someone book a table they are glad about, to remember them accurately, and never to act on their behalf without asking. Every axis on which Tree of Thought is strong, meaning autonomous exploration, internal evaluation, and commitment to a single searched out answer, is an axis on which this product deliberately does less.

6. The risk this decision accepts, and the mitigation

Choosing a fixed relaxation ladder over a search accepts one real risk. The ordering encodes a prior held by the designer, and that prior can be wrong for a particular guest. Someone for whom price is inviolable and cuisine is flexible will be served in the wrong order.

The mitigation is that the relaxation is announced rather than silent. The concierge tells the guest what it widened, for example that nothing vegan friendly was open in Midtown at that price on Friday, so it looked one band higher, and asks whether they would rather travel further instead. That hands the branch decision back to the person who holds the real preferences. It costs one conversational turn, no model overhead, and it follows the same principle that already governs the consent gate on standing preferences.

7. When this decision should be revisited

This is a judgement about this system, not a general position on Tree of Thought. It should be reopened if any of the following becomes true.

- **Candidate generation stops being retrieval.** An evening across several venues, with drinks, dinner and dessert, where travel time and closing hours interact, is a genuine combinatorial plan the model would have to construct. That is a problem shaped for a tree.
- **A trustworthy evaluator becomes available.** Enough booking outcomes and feedback gathered after dining would give pruning something real to stand on, extending the offline evaluation approach from retrieval to selection.
- **The product moves toward autonomy.** If the agent were ever asked to book without a human at the branch points, the search would have to move inside the system. At that point it should be a tree, evaluated and logged, rather than a single greedy chain.

Until then, the correct decision is to keep the breadth where the guest can see it.